

A/B TESTING

Module 3: Predicting the Future

1. THE BILLION DOLLAR EXPERIMENT

An A/B Test is the ultimate marriage of Statistics and Business. You take a live audience, split them into two random groups, and serve them two different versions of a feature (A = Control, B = Variant).

The goal is to mathematically prove that the change in Group B was not just a lucky fluctuation, but a genuine improvement.

2. THE TWO-SAMPLE Z-TEST

When we want to see if the “Conversion Rate” (the percentage of users who click a button) is statistically different between two groups, we use a **Two-Sample Z-Test**.

The Variables:

- p_A, p_B : The conversion rates (successes/total) of Group A and Group B.
- n_A, n_B : The total number of users in each group.
- $\hat{p}$: The pooled conversion rate: $\frac{\text{Total Successes}}{\text{Total Users}}$

The Z-Score Formula:

$$Z = \frac{(\hat{p}_B - \hat{p}_A)}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}$$

- If your calculated Z is greater than **1.96** (the critical value for a 95% confidence level), you have mathematically proven the difference is “statistically significant.”

3. THE SIGNIFICANCE THRESHOLD ($Z > 1.96$)

In the business world, we rarely look at P-values directly. We look at **Confidence Intervals**.

- **The logic:** If we are 95% confident that the true difference between Group B and Group A is greater than 0, we “ship” the feature.
- **The 1.96 Rule:** In a standard normal distribution, exactly 95% of the area under the curve falls between $Z = -1.96$ and $Z = +1.96$. If your Z-score pushes beyond that 1.96 mark, you are in the “extreme 2.5% tail,” which is the universal threshold for scientific discovery.

4. THE PITFALLS OF “PEEKING”

A massive mistake companies make is “**Peeking**”—checking the A/B test results every single hour and stopping the test the second the graph looks good.

- **Why it fails:** If you peek enough times, you are guaranteed to eventually see a “statistically significant” result even if the two groups are identical. This is a false positive trap.
- **The Solution:** Calculate your required sample size (N) *before* the test starts, and don’t stop the test until you hit that number.

5. FROM STATISTICS TO MACHINE LEARNING

A/B testing tells you *what* worked, but it doesn’t tell you *why* or how to predict future behavior. That is where we leave the world of simple averages and move into the world of **Linear Regression**, where we build actual mathematical models to predict the future.